

QUESTION 1 (15 Marks)

a) Let $z=5-i$ and $w= 3+2i$, Express the following in the form $a+ib$ where a,b are real numbers.

(i) $z + \bar{w}$ 1

(ii) $z + iw$ 1

b) Express $\frac{(1+3i)^2}{3+i}$ in the form $a+ib$ where a and b are real numbers 2

c) (i) Find the square roots of $16-30i$. Give your answers in the form $a+ib$ 2

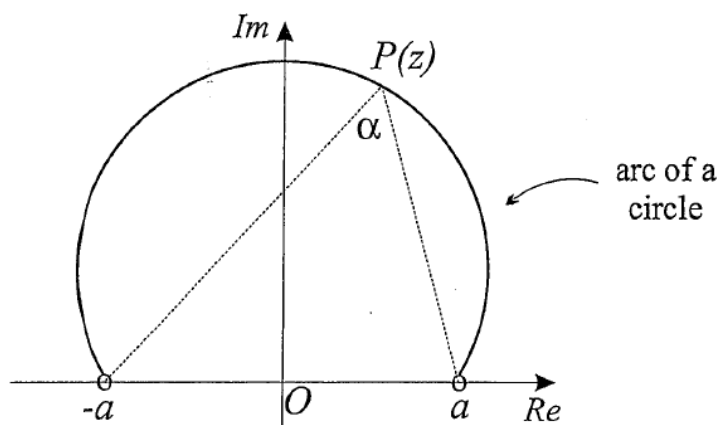
(ii) Hence solve the equation $z^2 - (3-i)z - 2 + 6i = 0$ 2

d) (i) Express $z = -3\sqrt{3} + 3i$ in mod-arg form 2

(ii) Hence find the smallest positive integer n such that z^n is real 1

e) Let $P(z)= z^3 + az^2 + bz + c$ where a,b and c are real.
Two of the roots of $P(z)=0$ are -2 and $(-3+2i)$. Find the value of a,b and c . 3

f)



In the diagram above, the locus of the point P representing the complex number z is graphed. Write down a possible equation in terms of z, b and α for the locus of P . Note that constants b and α are real.

QUESTION 2 (15 marks)

(a) On an Argand diagram sketch the locus of z satisfying

(i) $|z| = |z - 6 - 3i|$ 2

(ii) $\frac{1}{2}(z + \bar{z}) = |z| - 2$ 3

(b) The complex number z lies on the locus $\arg(z+i) = \frac{\pi}{4}$.

(i) Sketch the locus, showing any intercepts with the axes. 2

(ii) Find the least value of $|z|$ 2

(c) Let $z_1 = \frac{\sqrt{6} - i\sqrt{2}}{2}$ and $z_2 = 1 - i$

(i) Write z_1 and z_2 in mod-arg form 3

(ii) Show that $\frac{z_1}{z_2} = \cos \frac{\pi}{12} + i \sin \frac{\pi}{12}$ 1

(iii) Find the value of $\frac{z_1}{z_2}$ in the form $a+ib$ where a and b are in surd form.

Hence or otherwise find the exact surdic expression for $\cos \frac{\pi}{12}$ 2

QUESTION 3 (15 Marks)

(a)

(i) Expand and simplify $(\cos \theta + i \sin \theta)^5$ 2

(ii) Hence prove that $\cos 5\theta = 16 \cos^5 \theta - 20 \cos^3 \theta + 5 \cos \theta$ 2

(iii) Deduce that $\cos \frac{\pi}{10} \cos \frac{3\pi}{10} \cos \frac{7\pi}{10} \cos \frac{9\pi}{10} = \frac{5}{16}$ 2

(b) (i) Use De Moivre's Theorem to prove that, if $2 \cos \theta = x + \frac{1}{x}$ then

$$2 \cos n\theta = x^n + \frac{1}{x^n} \quad 1$$

(ii) Hence or otherwise, solve the equation

$$5x^4 - 11x^3 + 16x^2 - 11x + 5 = 0 \quad 4$$

(c) (i) Given that ω is one of the non-real roots of $z^3 = 1$ show that $1 + \omega + \omega^2 = 0$ 1

(ii) Using (i), or otherwise, show that

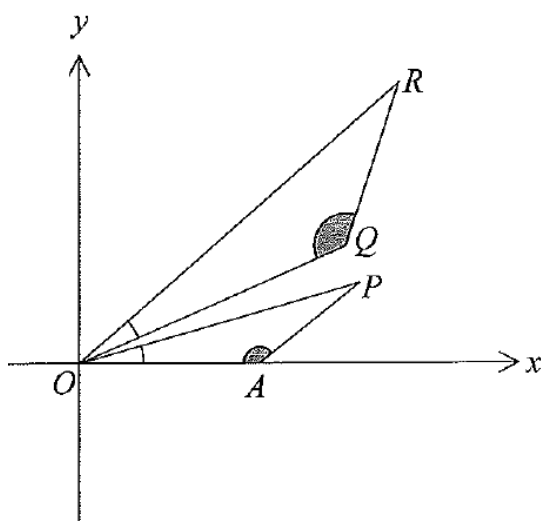
$$\left(\frac{\omega}{1 + \omega} \right)^k + \left(\frac{\omega^2}{1 + \omega^2} \right)^k = (-1)^k 2 \cos \frac{2}{3} k \pi \text{ where } k \in \mathbb{Z} \quad 3$$

QUESTION 4 (15 Marks)

- (a) Let z be a complex number of modulus 3 and w be a complex number of modulus 1

Show that $|z - w|^2 = 10 - (z\bar{w} + \bar{z}w)$ 2

- (b) In the figure below, the points P, Q and A represent the complex numbers z_1, z_2 and 1 respectively. By construction, $\angle OAP = \angle OQR$ and $\angle AOP = \angle QOR$. Explain why the point R represents the complex number $z_1 z_2$



3

- (c) A and B are two points in an Argand diagram representing the complex numbers $z_1 = -1$ and $z_2 = \cos \theta + i \sin \theta$ respectively where $\frac{\pi}{2} < \theta < \pi$.

C is the point representing the complex number $z_3 = z_1 + z_2$

- (i) Sketch the quadrilateral OACB on an Argand diagram where O is the point representing the complex 0.

Then mark an angle on the diagram which is equal to θ

3

- (ii) Let $z_4 = z_2 - z_1$

(α) Show that $\frac{z_4}{z_3} = i \left(\frac{\sin \theta}{\cos \theta - 1} \right)$, hence find $\arg \left(\frac{z_4}{z_3} \right)$ 4

- (β) Using α show that the diagonals of the quadrilateral OACB are perpendicular to each other. 3

MATHEMATICS Extension 2: Question 1.....

Suggested Solutions

Marks

Marker's Comments

(a) (i) $5-i + 3-2i = 8-3i$

①

(ii) $5-i + 3i-2 = 3+2i$

①

(b) $\frac{(1+3i)^2}{3+i} = \frac{1+6i-9}{3+i}$

$= \frac{-8+6i}{3+i} \times \frac{3-i}{3-i}$

①

$= \frac{-24+8i+18i+6}{9+1}$

$= \frac{-18}{10} + \frac{26i}{10}$

$= -\frac{9}{5} + \frac{13i}{5}$

①

(c) (i) $(x+iy)^2 = 16-30i$ $x, y \in \mathbb{R}$

$x^2-y^2+2xyi = 16-30i$

equate coefficients

$x^2-y^2 = 16$ --- (1)

$2xy = -30$ --- (2)

$y = -\frac{15}{x}$ Sub into (1)

$x^2 - \frac{225}{x^2} = 16$

$x^4 - 16x^2 - 225 = 0$

$(x^2-25)(x^2+9) = 0$

$x^2 = 25$ or $x^2 = -9$
 $x = \pm 5$ no real solution

so $y = -3$ or 3

square roots are $5-3i$, $-5+3i$

(ii) $z = \frac{(3-i) \pm \sqrt{(3-i)^2 - 4(1)(-2+6i)}}{2}$

$= \frac{(3-i) \pm \sqrt{9-6i-1+8-24i}}{2}$

①

many methods
-1 for process
-1 for answer

①

①

MATHEMATICS Extension 2: Question...1..... cont

Suggested Solutions

Marks

Marker's Comments

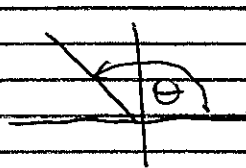
$$z = \frac{3-i \pm \sqrt{16-30i}}{2}$$

$$= \frac{3-i \pm (5+3i)}{2} \quad \text{from (1)}$$

$$= 4-2i \quad \text{or} \quad -1+i$$

①

(d) (i)



$$r = \sqrt{(3\sqrt{3})^2 + 3^2}$$

$$\therefore r = 6$$

$$\theta = \frac{5\pi}{6}$$

①

If your final answer is $6\text{cis}\frac{5\pi}{6}$ maximum is one mark.

①

$$-3\sqrt{3} + 3i = 6\left(\cos\frac{5\pi}{6} + i\sin\frac{5\pi}{6}\right)$$

$$(ii) \quad z^n = 6^n \left(\cos\frac{5\pi}{6} + i\sin\frac{5\pi}{6}\right)^n$$

$$= 6^n \left(\cos\frac{5n\pi}{6} + i\sin\frac{5n\pi}{6}\right) \quad (\text{De Moivre's Theorem})$$

$$\text{to be real then } \frac{\sin(5n\pi)}{6} = 0$$

$\frac{5n\pi}{6}$ must be ^{an integer} $\therefore n$ is a positive integer
e.g. $n=6$

①

right or wrong

(e) since the coefficients are all real the complex roots are conjugate pairs
so roots are $-2, -3+2i$ and $-3-2i$

①

$$\therefore (z+2)(z+(3-2i))(z+(3+2i))$$

$$= (z+2)(z^2+6z+13)$$

$$= z^3 + 8z^2 + 25z + 26$$

①

$$\therefore a=8, b=25, c=26$$

①

many ways of doing this question

$$\textcircled{f} \quad \arg\left(\frac{z-b}{z+b}\right) = \alpha$$

$$\text{or} \quad \arg(z-b) - \arg(z+b) = \alpha$$

①

right or wrong

$$\text{or} \quad \arg(z+b) - \arg(z-b) = -\alpha$$

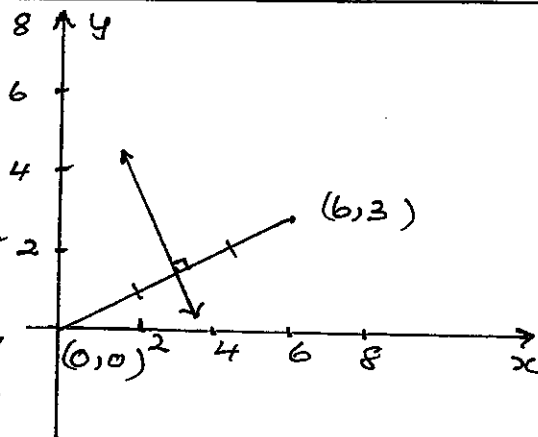
Q2

a(i)

$$|Z| = |Z - 6 - 3i|$$

$$|Z| = |Z - (6 + 3i)|$$

Locus is the perpendicular bisector of the interval joining $(0,0)$ and $(6,3)$



OR

$$|x + iy|$$

$$= |x + iy - 6 - 3i|$$

$$\sqrt{x^2 + y^2}$$

$$= \sqrt{(x-6)^2 + (y-3)^2}$$

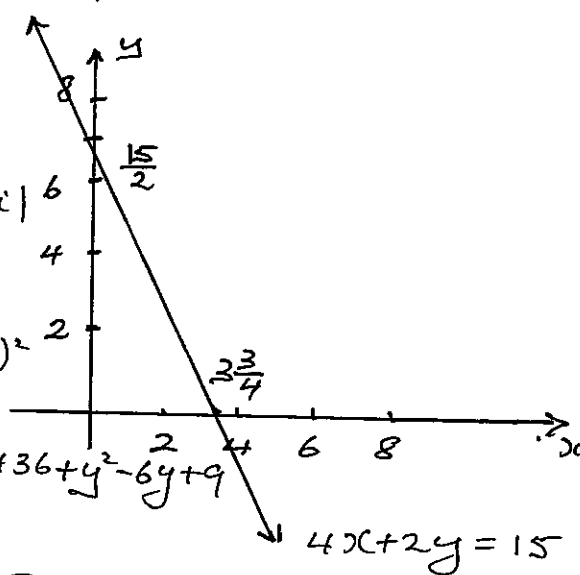
$$x^2 + y^2 = x^2 - 12x + 36 + y^2 - 6y + 9$$

$$12x + 6y = 45$$

$$4x + 2y = 15$$

$$x \text{ intercept } + 3\frac{3}{4}$$

$$y \text{ intercept } + 7\frac{1}{2}$$



2 Marks for correct diagram showing the geometrical significance of the locus

2 Marks for the correct equation of the locus and a correct diagram showing the x and y intercepts

1 mark for the correct equation

1 mark for a geometrical interpretation of the locus on a diagram with only one of
a) perpendicular
OR
b) bisector
shown

Note diagram drawn to scale

a)

(ii) $\frac{1}{2}(z + \bar{z}) = |z| - 2$

$$\frac{1}{2}(x + iy + x - iy) = \sqrt{x^2 + y^2} - 2$$

$$x = \sqrt{x^2 + y^2} - 2$$

$$x + 2 = \sqrt{x^2 + y^2}$$

$$(x + 2)^2 = x^2 + y^2$$

$$x^2 + 4x + 4 = x^2 + y^2$$

$$y^2 = 4x + 4$$

$$y^2 = 4(x + 1)$$

Parabola

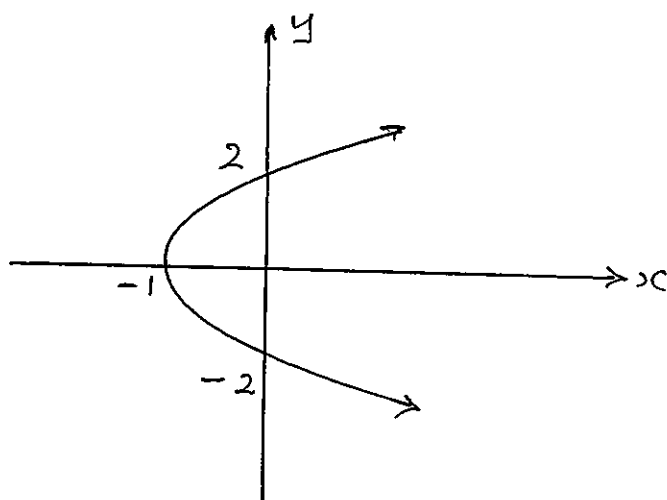
Vertex at $(-1, 0)$

Axis of symmetry

is the x axis

intercepts on y

axis of ± 2



3 Marks for correct Cartesian equation of the locus with a sketch showing the intercepts on the axes

2 Marks for a correct equation

1 Mark for an incorrect equation resulting from $\frac{1}{2}(x + iy + x - iy) = x^2 + y^2 - 2$

Some Candidates who wrote $y = \pm 2\sqrt{x+1}$ only drew the top half of the parabola.

b)

Equation of the line through $(0, -1)$ and $(1, 0)$

$$\text{is } \frac{x}{1} + \frac{y}{-1} = 1$$

$$\text{i.e. } x - y - 1 = 0$$

$$\begin{aligned} \text{least } |Z| &= \left| \frac{0 - 0 - 1}{\sqrt{1^2 + (-1)^2}} \right| \\ &= \frac{1}{\sqrt{2}} \end{aligned}$$

OR

$$AB^2 = 1^2 + 1^2$$

$$= 2$$

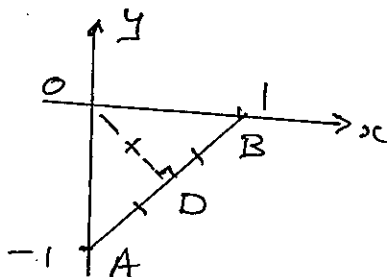
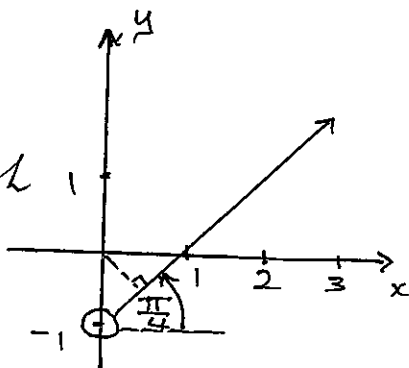
$$AB = \sqrt{2}$$

$$OD = \frac{1}{2} AB$$

$$= \frac{1}{2} \times \sqrt{2}$$

$$= \frac{1}{\sqrt{2}}$$

$$\therefore \text{least } |Z| = \frac{1}{\sqrt{2}}$$



2 Marks for correct sketch showing open circle at $(0, -1)$ and a ray through $(1, 0)$

1 Mark for correct sketch without arrow on ray

1 Mark for correct sketch without open circle

Ray must be drawn using a ruler

2

Least value of $|Z|$ was found using perpendicular distance from $(0, 0)$ to $x - y - 1 = 0$

OR

Using Pythagoras Theorem

$$c) (i) |z_1| = \sqrt{\left(\frac{\sqrt{6}}{2}\right)^2 + \left(\frac{-\sqrt{2}}{2}\right)^2} \quad |z_2| = \sqrt{1^2 + (-1)^2}$$

$$= \sqrt{\frac{6}{4} + \frac{2}{4}} \quad = \sqrt{2}$$

$$= \sqrt{2}$$

$$\text{Arg } z_1 = \tan^{-1}\left(\frac{-\sqrt{2}}{\sqrt{6}}\right) \quad \text{Arg } z_2 = \tan^{-1}\left(\frac{-1}{1}\right)$$

$$= \tan^{-1}\left(\frac{-1}{\sqrt{3}}\right) \quad = \tan^{-1}(-1)$$

$$= -\frac{\pi}{6} \quad = -\frac{\pi}{4}$$

$$z_1 = \sqrt{2} \left(\cos\left(-\frac{\pi}{6}\right) + i \sin\left(-\frac{\pi}{6}\right) \right)$$

$$z_2 = \sqrt{2} \left(\cos\left(-\frac{\pi}{4}\right) + i \sin\left(-\frac{\pi}{4}\right) \right)$$

$$\text{Let } \text{Cis } \theta = \cos \theta + i \sin \theta$$

$$\frac{z_1}{z_2} = \frac{\sqrt{2} \text{Cis}\left(-\frac{\pi}{6}\right)}{\sqrt{2} \text{Cis}\left(-\frac{\pi}{4}\right)}$$

$$= \text{Cis}\left(-\frac{\pi}{6} - -\frac{\pi}{4}\right)$$

$$= \text{Cis}\left(-\frac{2\pi}{12} + \frac{3\pi}{12}\right)$$

$$= \text{Cis } \frac{\pi}{12}$$

$$= \cos \frac{\pi}{12} + i \sin \frac{\pi}{12}$$

$$\frac{z_1}{z_2} = \frac{\frac{\sqrt{6}}{2} - i \frac{\sqrt{2}}{2}}{1 - i}$$

$$= \frac{\sqrt{6} - i\sqrt{2}}{2(1-i)} \times \frac{(1+i)}{(1+i)}$$

$$= \frac{\sqrt{6} + \sqrt{6}i - i\sqrt{2} - i^2\sqrt{2}}{2(1-i^2)}$$

$$= \frac{\sqrt{6} + \sqrt{2} + i\sqrt{6} - \sqrt{2}}{4}$$

$$= \cos \frac{\pi}{12} + i \sin \frac{\pi}{12}$$

$$\text{Equating Real Parts } \cos \frac{\pi}{12} = \frac{\sqrt{6} + \sqrt{2}}{4}$$

1 Mark for both $|z_1|$ and $|z_2|$ correct

1 Mark for Arg z_1

1 Mark for Arg z_2

3 Marks only if both z_1 and z_2 expressed in mod-arg form.

1 Mark for

$$\text{Cis}\left(-\frac{\pi}{6} - -\frac{\pi}{4}\right)$$

$$= \text{Cis}\left(\frac{\pi}{12}\right)$$

0 Marks for

$$\frac{\text{Cis } -\frac{\pi}{6}}{\text{Cis } -\frac{\pi}{4}} = \text{Cis } \frac{\pi}{12}$$

MUST SHOW

Otherwise Approach

$$\cos \frac{\pi}{12} = \cos\left(\frac{\pi}{4} - \frac{\pi}{6}\right)$$

OR

$$\cos 2A = 2\cos^2 A - 1$$

$$\cos \frac{\pi}{6} = 2\cos^2 \frac{\pi}{12} - 1$$

Suggested Solutions

Marks

Marker's Comments

a) i) Using Binomial Expansion:

$$(\cos \theta + i \sin \theta)^5 = \cos^5 \theta + 5i \cos^4 \theta \sin \theta + 10i^2 \cos^3 \theta \sin^2 \theta + 10i^3 \cos^2 \theta \sin^3 \theta + 5i^4 \cos \theta \sin^4 \theta + i^5 \sin^5 \theta$$

$$= (\cos^5 \theta - 10 \cos^3 \theta \sin^2 \theta + 5 \cos \theta \sin^4 \theta) + i(5 \cos^4 \theta \sin \theta - 10 \cos^2 \theta \sin^3 \theta + \sin^5 \theta)$$

1

Also, by de Moivre's Theorem

$$(\cos \theta + i \sin \theta)^5 = \cos 5\theta + i \sin 5\theta$$

1

Words (de Moivre) necessary for mark.

ii) Equating real parts of each expansion

$$\begin{aligned} \cos 5\theta &= \cos^5 \theta - 10 \cos^3 \theta \sin^2 \theta + 5 \cos \theta \sin^4 \theta \\ &= \cos^5 \theta - 10 \cos^3 \theta (1 - \cos^2 \theta) + 5 \cos \theta (1 - \cos^2 \theta)^2 \\ &= \cos^5 \theta - 10 \cos^3 \theta + 10 \cos^5 \theta + 5 \cos \theta - 10 \cos^3 \theta + 5 \cos^5 \theta \\ &= 16 \cos^5 \theta - 20 \cos^3 \theta + 5 \cos \theta \end{aligned}$$

1

for this far and the wording

iii) The 5 zeroes of this expression will be 5 distinct zeroes of

$$\cos 5\theta = 0$$

$$\cos \theta \text{ when } 5\theta = \frac{\pi}{2}, \frac{3\pi}{2}, \frac{5\pi}{2}, \frac{7\pi}{2}, \frac{9\pi}{2}$$

$$\text{i.e. } \cos \theta \text{ when } \theta = \frac{\pi}{10}, \frac{3\pi}{10}, \frac{5\pi}{10}, \frac{7\pi}{10}, \frac{9\pi}{10}$$

Factoring out the zero $\cos \theta = 0$ leaves a polynomial of degree 4 with zeroes $\cos \frac{\pi}{10}, \cos \frac{3\pi}{10}, \cos \frac{7\pi}{10}, \cos \frac{9\pi}{10}$

1

Suggested Solutions

Marks

Marker's Comments

The product of the roots = " e/a " = $5/16$

$$\therefore \frac{\cos \frac{\pi}{10}}{10} \frac{\cos \frac{3\pi}{10}}{10} \frac{\cos \frac{7\pi}{10}}{10} \frac{\cos \frac{9\pi}{10}}{10} = \frac{5}{16}$$

1

[An alternative is to consider the sum of roots taken 4 at a time from the degree 5 equation. As the root 0 occurs in all but one of the products, we arrive at the same result above.]

b) i) $2 \cos \theta = x + \frac{1}{x}$

$$x^2 - 2x \cos \theta + 1 = 0$$

$$\therefore x = \frac{2 \cos \theta \pm \sqrt{4 \cos^2 \theta - 4}}{2}$$

$$\therefore x = \cos \theta \pm i \sin \theta$$

$$= \cos(\pm \theta) + i \sin(\pm \theta)$$

$$\therefore x^n = \cos(\pm n\theta) + i \sin(\pm n\theta) \quad (\text{de Moivre})$$

$$x^{-n} = \cos(\mp n\theta) + i \sin(\mp n\theta) \quad (\text{" "})$$

$$\frac{x^n + \frac{1}{x^n}}{x^n} = 2 \cos n\theta \quad (\text{Adding above lines})$$

(Noting that $\cos(-x) = \cos x$, $\sin(-x) = -\sin x$)

ii) Divide by x^2 ($x \neq 0$)

$$5\left(x^2 + \frac{1}{x^2}\right) - 11\left(x + \frac{1}{x}\right) + 16 = 0$$

If $x = \cos \theta + i \sin \theta$, using part (i);

$$10 \cos 2\theta - 22 \cos \theta + 16 = 0$$

It was allowed (although not strictly correct) if the statement $x = \cos \theta + i \sin \theta$ was made without proof.

1

1

Suggested Solutions	Marks	Marker's Comments
Putting $\cos 2\theta = 2\cos^2\theta - 1$, $20\cos^2\theta - 22\cos\theta + 6 = 0$ $10\cos^2\theta - 11\cos\theta + 3 = 0$ $(5\cos\theta - 3)(2\cos\theta - 1) = 0$ $\cos\theta = \frac{3}{5} \text{ or } \frac{1}{2}$ $\therefore \sin\theta = \pm \frac{4}{5} \text{ or } \pm \frac{\sqrt{3}}{2}$ $x = \cos\theta + i\sin\theta$ $= \frac{3 \pm 4i}{5} \text{ or } \frac{1 \pm i\sqrt{3}}{2}$ [An alternative is to modify the original equation to $5\left(x + \frac{1}{x}\right)^2 - 11\left(x + \frac{1}{x}\right) + 6 = 0$ and solve for $\left(x + \frac{1}{x}\right)$. It is then necessary to solve a quadratic equation for each root, giving the same 4 roots of x.]	1 1	
c) i) If ω is a root of $z^3 = 1$, then $\omega^3 = 1$ i.e. $\omega^3 - 1 = 0$ $\therefore (\omega - 1)(\omega^2 + \omega + 1) = 0$ But $\omega \neq 1$ since ω is non real $\therefore \underline{\omega^2 + \omega + 1 = 0}$	1	There was a lot of untidiness with z. Substitute ω early as shown here.

Suggested Solutions

Marks

Marker's Comments

c) i) (Alternative)

If ω is a non real root, then ω^2 is also a root.

$$(\omega^2)^3 = (\omega^3)^2 = 1$$

It is a different root

$$\omega^2 - \omega = \omega(\omega - 1) \neq 0$$

And 1 is clearly a real root.

Sum of the roots $\underline{\underline{1 + \omega + \omega^2 = -\frac{1}{2} = 0}}$

ii) $1 + \omega = -\omega^2$

$$1 + \omega^2 = -\omega$$

$$\therefore \frac{\omega}{1 + \omega} = \frac{\omega}{-\omega^2}$$

$$\frac{\omega^2}{1 + \omega^2} = \frac{\omega^2}{-\omega}$$

$$= -\frac{1}{\omega}$$

$$= -\omega$$

$$\therefore \left(\frac{\omega}{1 + \omega}\right)^k + \left(\frac{\omega^2}{1 + \omega^2}\right)^k = \left(-\frac{1}{\omega}\right)^k + (-\omega)^k$$

$$= (-1)^k \left(\frac{1}{\omega^k} + \omega^k\right)$$

But from (b) i) since $\omega = \cos \frac{2\pi}{3}$,

$$\omega + \frac{1}{\omega} = 2 \cos \frac{2\pi}{3}$$

$$\therefore \omega^k + \frac{1}{\omega^k} = 2 \cos \frac{2k\pi}{3}$$

$$\therefore \left(\frac{\omega}{1 + \omega}\right)^k + \left(\frac{\omega^2}{1 + \omega^2}\right)^k = (-1)^k 2 \cos \frac{2k\pi}{3}$$

[If $\omega = \cos \frac{4\pi}{3}$ had been used, the same result would occur since $\cos \frac{4k\pi}{3} = \cos \frac{2k\pi}{3}$ for any $k \in \mathbb{Z}$]

There were a lot of errors and poor explanation occurring when people solved $z^3 = 1$ and used k as a parameter.

This was to be avoided at all costs.

MATHEMATICS Extension 2: Question 4

Suggested Solutions

Marks

Marker's Comments

a) $|z| = 3$ $|w| = 1$

$$|z-w|^2 = (z-w)(\overline{z-w})$$

$$= (\overline{z-w})(\overline{z-w})$$

$$= \overline{z}\overline{z} - \overline{z}\overline{w} - \overline{w}\overline{z} + \overline{w}\overline{w}$$

$$= |z|^2 + |w|^2 - (\overline{z}\overline{w} + \overline{w}\overline{z})$$

$$= 9 + 1 - (\overline{z}\overline{w} + \overline{w}\overline{z})$$

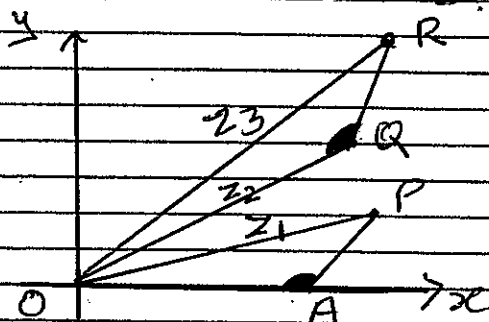
$$= 10 - (\overline{z}\overline{w} + \overline{w}\overline{z})$$

①

①

Accepted many methods involving manipulations of LHS and RHS. Did not accept "special cases"

(b) Let R represent complex number z_3 .



In ΔORQ and ΔOPA

$\angle OAP = \angle ORQ$ (Given)

$\angle POA = \angle ROQ$ (Given)

$\therefore \Delta ORQ \sim \Delta OPA$ (equiangular)

$$\therefore \frac{OA}{OQ} = \frac{OP}{OR}$$

Corresponding sides in similar triangles are in the same ratio

$$\frac{1}{|z_2|} = \frac{|z_1|}{|z_3|}$$

$$|z_3| = |z_1| |z_2|$$

$$= |z_1 \times z_2|$$

$$\arg z_1 + \arg z_2 = \angle AOP + \angle POQ$$

$$= \angle AOP + \angle POQ + \angle QOR$$

$$= \angle AOR$$

$$= \arg z_3$$

$$|z_3| = |z_1 \times z_2|$$

$$\arg z_3 = \arg z_1 + \arg z_2$$

$$= \arg(z_1 \cdot z_2)$$

$$\therefore z_3 = z_1 \cdot z_2$$

①

①

①

This was a "show" question

Must give complete proof for similar triangles with reasons

ratio of moduli with full reason

Must show args in terms of angles

No loss if conclusion not written

MATHEMATICS Extension 1: Question 4

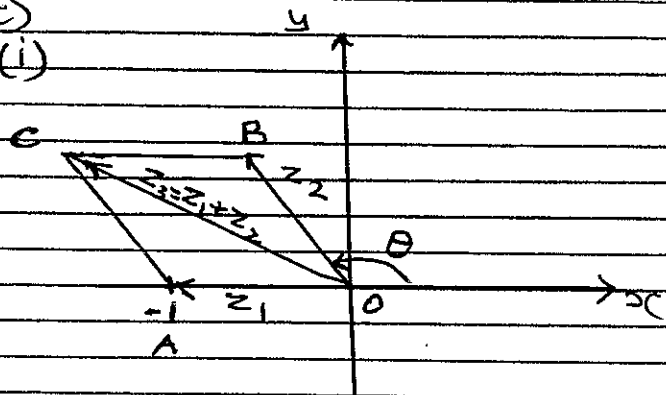
Suggested Solutions

Marks

Marker's Comments

(c)

(i)



①

 θ is obtuse

①

$$|z_1| = |z_2| = 1$$

①

Must show points A, B, C correctly.

$$(ii) \quad z_4 = (\cos \theta + i) + i \sin \theta$$

$$(a) \quad z_3 = -1 + \cos \theta + i \sin \theta = (\cos \theta - 1) + i \sin \theta$$

$$\frac{z_4}{z_3} = \frac{(\cos \theta + 1) + i \sin \theta}{(\cos \theta - 1) + i \sin \theta} \times \frac{(\cos \theta - 1 - i \sin \theta)}{(\cos \theta - 1 - i \sin \theta)}$$

$$= \frac{\cos^2 \theta - 1 - i \sin \theta (\cos \theta + 1) + i \sin \theta (\cos \theta - 1) + \sin^2 \theta}{(\cos \theta - 1)^2 + \sin^2 \theta}$$

$$= \frac{-1 - 1 - i \sin \theta \cos \theta - i \sin \theta + i \sin \theta \cos \theta - i \sin \theta}{\cos^2 \theta - 2 \cos \theta + 1 + \sin^2 \theta}$$

$$= \frac{-2i \sin \theta}{2 - 2 \cos \theta}$$

$$= \frac{i \sin \theta}{\cos \theta - 1}$$

$\frac{z_4}{z_3}$ is purely imaginary

$$\therefore \frac{\pi}{2} < \theta < \pi \quad (\text{2nd Quadrant})$$

$$\therefore \sin \theta > 0$$

$$\cos \theta - 1 < 0$$

$$\therefore \frac{\sin \theta}{\cos \theta - 1} < 0$$

$$\therefore \arg \left[\frac{z_4}{z_3} \right] = -\frac{\pi}{2}$$

$$(b) \quad \arg \left[\frac{z_4}{z_3} \right] = -\frac{\pi}{2}$$

$$\arg z_4 - \arg z_3 = -\frac{\pi}{2}$$

$$\arg z_3 = \arg z_4 + \frac{\pi}{2}$$

$\therefore \frac{\pi}{2}$ is angle between z_3 and z_4

$$z_3 = OC \text{ and } z_4 = AB$$

are diagonals of OACB

$\therefore$ diagonals are perpendicular

OTHER METHODS POSSIBLE

BUT MUST REFER to part (a)

① Realizing

$$\frac{z_4}{z_3}$$

unsimplified

① correctly simplifying

① purely imaginary

① show for $\frac{\pi}{2} < \theta < \pi$
 $\frac{\sin \theta}{\cos \theta - 1} < 0$ ① $\frac{\pi}{2}$.① difference in args is $\frac{\pi}{2}$ ① z_3, z_4 are diagonals.